Swati Aggrawal

swatiagg357@gmail.com | github.com/swatiaggrawal |linkedin.com/in/swatiaggrawal

PROFESSIONAL SUMMARY

MSc Computer Science student at Sapienza Università di Roma specializing in Machine Learning and Artificial Intelligence. Experienced in developing machine learning solutions, evaluating model performance, analyzing complex datasets, and deploying ML applications using Python, FastAPI, Docker, and AWS. Strong background in NLP, Computer Vision, feature engineering, statistical analysis, and model evaluation. Published IEEE researcher with hands-on experience translating machine learning concepts into practical, data-driven solutions. Seeking opportunities in Machine Learning Engineering, Data Science, and AI Consulting.

EDUCATION

Sapienza Università di Roma **May 2026**

Master's in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

SKILLS

- **Programming :** Python, Bash, SQL, C++
- **Machine Learning :** Supervised Learning, Feature Engineering, Model Evaluation, Classification, Regression, CNNs, Transformers, Continual Learning
- **Data Analysis :** Statistical Analysis, Data Cleaning, Exploratory Data Analysis, Performance Evaluation
- **Machine Learning Frameworks :** PyTorch, TensorFlow, Keras, Sentence Transformers
- **Deployment & Cloud :** FastAPI, REST APIs, Docker, AWS ECS, AWS Fargate, CloudWatch
- **Tools :** Git, GitHub, Linux

RESEARCH EXPERIENCE

Continual Learning for Computer Vision **May 2025**

- **Conducted a structured review of 60+ papers** on catastrophic forgetting, replay/regularisation methods, and TIL/DIL/CIL protocols.
- Shaped thesis research and experimental design by identifying open challenges in continual learning, guiding novel experiments to address key gaps.

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- **Published IEEE ViTECoN-2023 survey paper** reviewing 6 ML models (RF, SVM, GA, LS-SVM, Decision Trees, ANN) for air quality prediction across 6 NAAQS pollutants.
- Analysed prediction-based and knowledge discovery paradigms, identified real-time monitoring and IoT integration as priority research gaps and **cited 6 times** since publication.

PROJECTS

DocDelta [Document Comparison API]

- Developed an end-to-end NLP solution for semantic comparison of legal and contract documents using Sentence Transformers and cosine similarity.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.

- Evaluated inference performance and reduced latency bottlenecks through model preloading and optimization techniques, achieving approximately 840ms warm inference latency.

Lossy Text Compressor [Semantic NLP Pipeline]

- Developed a machine learning pipeline using GTR-T5 embeddings, vector quantization, and text reconstruction to analyze compression-quality trade-offs.
- Performed systematic experiments across multiple quantization levels and evaluated results using cosine similarity and ROUGE-L metrics.
- Achieved optimal performance at 6-bit quantization with cosine similarity of 0.964 and ROUGE-L score of 0.776.
- Identified non-linear relationships between compression levels and reconstruction quality, providing insights for efficient semantic storage and retrieval.